

- Binary constant δ_{ij} where

$$\delta_{ij} = \begin{cases} 1 & \text{if station } s_i \text{ is covered by an AP at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

$$X_1 = 1$$

$$X_2 = 0$$

$$X_3 = 1$$

$$X_4 = 0$$

How many APs
are used? 2

$$X_1 = 1$$

$$X_2 = 0$$

$$X_3 = 1$$

$$X_4 = 1$$

How many APs
are used? 3

Can you find an expression of the
number of access points used in terms of
 x_1, x_2, x_3 and x_4 ?

$$X_1 + x_2 + x_3 + x_4$$

Decision variables:

$$x_j = \begin{cases} 1 & \text{if an AP is to be installed at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

Integer programming formulation:

$$\min \boxed{\text{The number of access points (An expression in } x_j)}$$

subject to

$$\boxed{\text{Each station is covered (One expression for each station, need } x_j \text{ and } \delta_{i,j})}$$

$$x_j \in \{0, 1\}$$

Question: What is the interpretation of
($p = 2$)

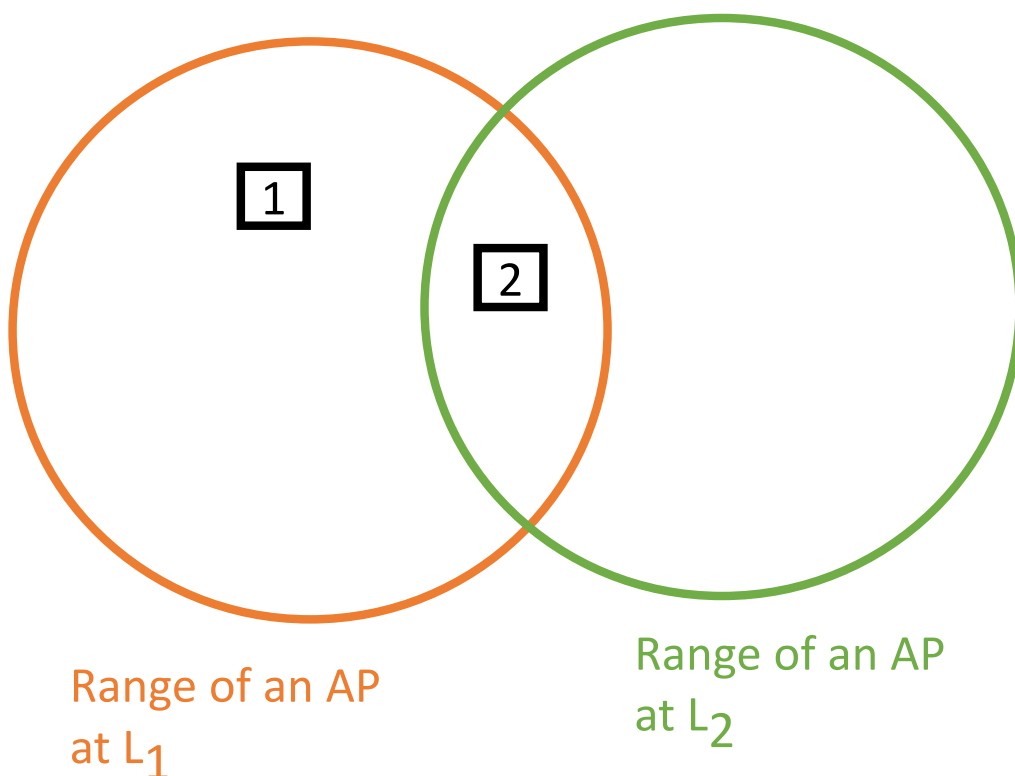
$$\sum_{j=1}^p \delta_{ij} x_j$$

■ Binary constant δ_{ij} where

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Assuming $x_1 = 1$ and $x_2 = 1$, fill in the last column

	Location 1	Location 2	$\delta_{i1} x_1 + \delta_{i2} x_2$
Station 1	$\delta_{11} = 1$	$\delta_{12} = 0$	$1*1+0*1=1$
Station 2	$\delta_{21} = 1$	$\delta_{22} = 1$	$1*1+1*1=2$



Question: What is the interpretation of $\sum_{j=1}^p \delta_{ij} x_j$ (p = 2)

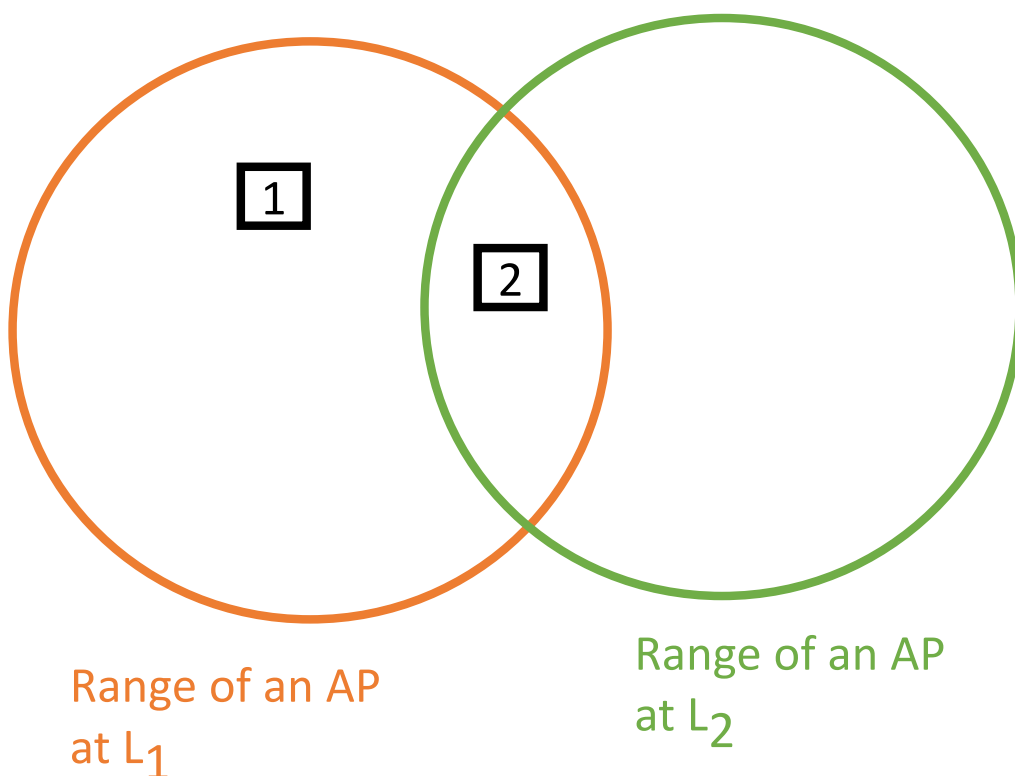
$$\sum_{j=1}^p \delta_{ij} x_j$$

■ Binary constant δ_{ij} where

$$\delta_{ij} = \begin{cases} 1 & \text{if station } s_i \text{ is covered by an AP at location } L_j \\ 0 & \text{otherwise} \end{cases}$$

Assuming $x_1 = 0$ and $x_2 = 1$, fill in the last column

	Location 1	Location 2	$\delta_{i1} x_1 + \delta_{i2} x_2$
Station 1	$\delta_{11} = 1$	$\delta_{12} = 0$	0
Station 2	$\delta_{21} = 1$	$\delta_{22} = 1$	1



$$x = 155y_1 + 466y_2 + 622y_3$$

$$y_1 + y_2 + y_3 = 1$$

$$y_1, y_2, y_3 \in \{0, 1\}$$

What are the combinations of y_1 , y_2 and y_3 that can satisfy the constraints in the box?

No: $(1,1,1)$, $(0,0,0)$, $(0,1,1)$

Yes: $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ no more

- A Cloud computing service provider offers 3 different packages with different speed and cost for each package. You can buy any cycles from any package but the deal requires that
 - # cycles from Package 1 + # cycles from Package 2 ≥ 10000 , or,
 - # cycles from Package 2 + # cycles from Package 3 ≥ 50000
 - At least one of these two inequalities must hold, but not necessarily both
- Let w_i = number of cycles to be bought from Package i

Case 1: $x_1 + x_2 > 1$

$$1 - p \geq x_1 + x_2 - 1 > 0$$

$$\Rightarrow 1 - p > 0 \Rightarrow p = 0$$

$$\Rightarrow -y + 4 \leq 4 * 0 = 0$$

$$\Rightarrow y \geq 4$$

If-then constraints

- We may want to impose if-then constraints, e.g.

$$\text{if } x_1 + x_2 > 1, \text{ then } y \geq 4$$

where x_1, x_2 are binary variables, and $0 \leq y \leq 10$

- The above if-then constraint can be captured by using an additional binary decision variable p

$$x_1 + x_2 - 1 \leq 1 - p$$

$$-y + 4 \leq 4p$$

$$p \in \{0, 1\}$$

Case 2: $x_1 + x_2 \leq 1$

$$1 - p \geq x_1 + x_2 - 1 \text{ (0 or -ve)}$$

p can be 0 or 1

$$\text{If } p = 0, -y + 4 \leq 0 \Rightarrow y \geq 4$$

$$\text{If } p = 1, -y + 4 \leq 4 \Rightarrow y \geq 0$$

$$y \geq 0$$

If-then constraints

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$$\begin{aligned} x_1 + x_2 - 1 &\leq 1 - p \\ -y + 4 &\leq 4p \\ p &\in \{0, 1\} \end{aligned}$$

$$f(x_1, x_2) = x_1 + x_2 - 1$$

$$g(y) = y - 4$$

$$x_1 + x_2 - 1 \leq M_1(1 - p)$$

$$4 - y \leq M_2 p$$

Find M_1 and M_2 such that:

$$x_1 + x_2 - 1 \leq M_1 \quad , \quad 4 - y \leq M_2$$

- Example: A Cloud computing service provider may use a progressive charging scheme
 - 5 dollars/sec for the first 5,000 sec
 - 2 dollars/sec for the next 15,000 sec
 - 1 dollar/sec thereafter

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■ Decision variables

$$y_i = \begin{cases} 1 & \text{if segment } i \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

- Segment 1: $0 \leq t \leq 5000$, cost = $5t$
- Segment 2: $5000 \leq t \leq 20000$, cost = $2t + 15000$
- Segment 3: $20000 \leq t$, cost = $t + 35000$

■ We can rewrite these as

■ $0 \leq ty_1 \leq 5000y_1$

■ $5000y_2 \leq ty_2 \leq 20000y_2$

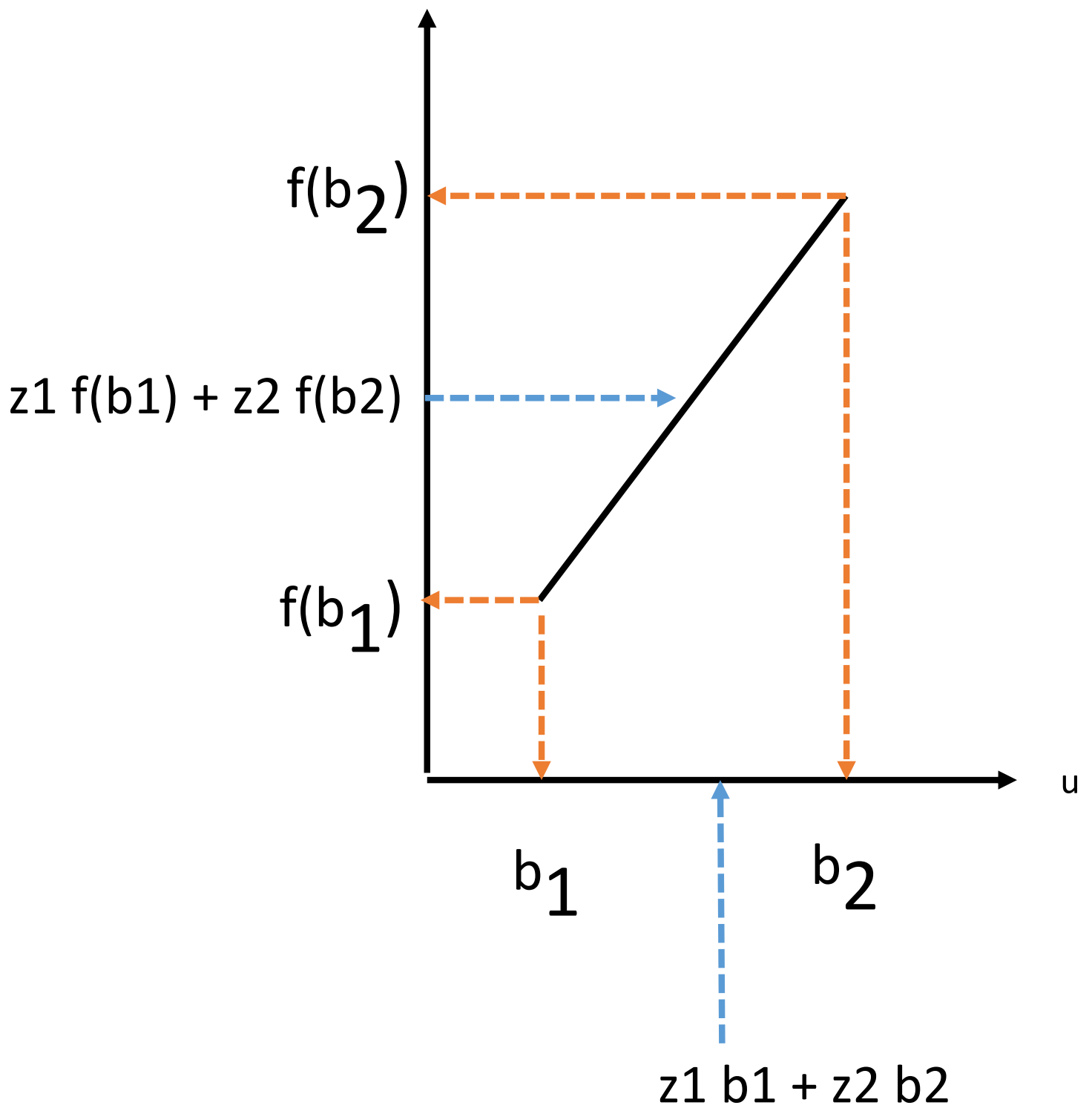
■ $20000y_3 \leq ty_3$

■ $\text{cost} = y_1(5t) + y_2(2t + 15000) + y_3(t + 35000)$

■ $y_1 + y_2 + y_3 = 1$

■ Problem: non-linear constraints

$$\begin{aligned} t_1 + t_2 + t_3 &= t y_1 + t y_2 + t * y_3 \\ &= t (y_1 + y_2 + y_3) \\ &= t \end{aligned}$$



$$y_1 = 1, y_2 = 0, y_3 = 0 \Rightarrow z_3, z_4 = 0$$

$$y_2 = 1, y_1 = 0, y_3 = 0 \Rightarrow z_1, z_4 = 0$$

Response time

